Comprehensive Analysis of Spine-Focused Fluoroscopic and Radiographic Datasets for Image-Guided Robotic Interventions

Introduction to Robotic Navigation and Fluoroscopic Imaging Methodologies

The evolution of robotic systems for endovascular and orthopedic interventions relies fundamentally on the availability of robust, highly annotated imaging datasets. In the context of transcatheter heart valve interventions, bi-plane fluoroscopy serves as the primary modality for spatial navigation, device tracking, and procedural evaluation. The reference dataset utilized for robotic cardiac guide sheath experiments—such as those evaluating dual-bendable, compressible robotic sheaths for safe cardiac delivery—comprises 800×800 pixel fluoroscopic X-ray images. These images are captured from paired left and right perspectives to enable epipolar-constrained three-dimensional (3D) reconstruction of large-diameter catheters crossing the fossa ovalis within a heart model phantom. Supported by dense mask annotations, this data facilitates the quantitative benchmarking of catheter tip localization and full shape detection performance.

Transitioning from the cardiovascular domain to the spinal domain requires identifying imaging datasets that mirror these highly specific technical characteristics. Specifically, equivalent datasets must feature dynamic temporal sequences, multi-view or bi-plane geometries, robust mask or keypoint annotations, and scenarios involving instrumentation. While cardiovascular interventions focus on navigating flexible sheaths through soft tissue cavities, spinal interventions involve rigid or semi-rigid instrumentation—such as pedicle screws, surgical drills, and robotic alignment modules—interacting with complex, overlapping osseous structures. Consequently, fluoroscopic spine datasets must address the inherent challenges of high-density bone overlap, tissue deformation, vertebral axial rotation, and the necessity for sub-millimeter tracking accuracy.

This report presents an exhaustive, highly technical analysis of available fluoroscopic, radiographic, and simulated X-ray datasets featuring the spine. It deconstructs their acquisition methodologies, annotation frameworks, and applicability to machine learning pipelines for robotic navigation, epipolar-constrained 3D reconstruction, and dynamic instrument tracking.

The Methodological Blueprint: Mapping Cardiac Navigation Requirements to Spinal Counterparts

To satisfy the parameters established by robotic cardiac sheath delivery paradigms, a comparable spinal dataset must exhibit several highly specific architectural traits. The

cardiovascular dataset relies on continuous X-ray sequences to capture the temporal kinematics of the robotic sheath as it is controlled via joystick by a clinician. Similarly, spinal datasets must capture either the dynamic kinematics of the spine itself or the real-time trajectory of surgical instruments.

The reliance on paired left and right perspective image subsets in the reference cardiac dataset is specifically engineered to facilitate epipolar-constrained 3D reconstruction. In a single two-dimensional (2D) X-ray projection, depth information is irrevocably lost, as all structures along the X-ray beam path are superimposed onto a single plane. However, when two images are captured simultaneously from known, distinct angles, any point in 3D space projects onto mathematically predictable epipolar lines on the 2D detectors. In spinal applications, similar stereotactic or multi-view paradigms are absolutely essential. Due to the complex morphology of vertebrae—including the pedicles, lamina, and spinous processes—a single anterior-posterior (AP) or lateral fluoroscopic projection is insufficient for depth estimation or safe surgical navigation. Therefore, bi-plane fluoroscopy, dual-source C-arm setups, or sparse multi-view acquisitions represent the equivalent imaging topology[1].

Furthermore, the cardiovascular dataset provides mask annotations for full shape detection of the catheter. In the spinal domain, annotations vary across a spectrum of complexity. Bounding boxes are frequently used for the gross localization of vertebrae or surgical implants[4]. Keypoint annotations, or landmarks, are utilized to identify vertebral corners, centroids, or pedicle entry points for pose estimation[6]. Finally, pixel-wise segmentation masks serve as the direct equivalent to the reference dataset, isolating individual vertebrae, the spinal canal, or surgical instruments from the background tissue and image noise[2].

Experimental Phantom Design and Bi-Plane Robotic Setup

Before analyzing specific digital datasets, it is critical to examine the physical experimental setups used to generate them, as this directly impacts dataset utility. In the cardiac studies, the robotic system steered a guide sheath across the fossa ovalis within a heart model phantom. The generation of equivalent spinal datasets relies on highly sophisticated patient-specific phantoms and cadaveric models.

To create realistic patient-specific phantoms for bi-plane X-ray evaluation, researchers utilize a multi-step digital and physical manufacturing pipeline. Initially, end-diastolic cardiac or spinal computed tomography (CT) scans are imported in the Digital Imaging and Communications in Medicine (DICOM) format into 3D image processing software, such as Materialize Mimics[9]. Specific Hounsfield Unit thresholds are applied to segment the target anatomy, enabling a 3D representation of the structures in a single mask. The 3D segmentation is exported as a Standard Tessellation Language (STL) file and processed in software like Geomagic Wrap to trim unnecessary artifacts, smooth the meshwork, and apply a water-tight shell thickness[9]. Support structures are added using computer-aided design (CAD) software to fix the relative distances between anatomies[9].

The physical models are then printed using high-resolution systems, such as the Stratasys Object Connex 260, utilizing rigid and translucent materials like VeroClear[9]. To conduct mock

catheterization or spinal drilling procedures under a bi-plane C-arm X-ray machine, the phantom is often integrated into an acrylic enclosure filled with water[9]. The water immersion matches the radiodensity of surrounding soft tissues, thereby eliminating unnatural air-contrast artifacts and ensuring that the resulting fluoroscopic images accurately represent clinical conditions[9]. This rigorous physical setup is the foundation upon which datasets like Veriserum and X23D are built.

Dynamic Fluoroscopic Video Datasets of the Spine

While static X-rays are abundant in medical literature, continuous fluoroscopic video datasets containing the spine are exceptionally rare due to stringent radiation exposure limits and patient privacy constraints. However, specific functional imaging datasets provide the temporal sequence data necessary for training advanced tracking algorithms.

The VFSS-5k Videofluoroscopy Dataset

The Videofluoroscopic Swallow Study (VFSS) dataset, known as VFSS-5k, represents one of the most comprehensive dynamic fluoroscopy datasets available featuring spinal anatomy[10]. Although primarily designed for dysphagia assessment—specifically tracking bolus trajectory, muscle kinematics, and hyoid bone movement—the cervical spine is continuously visualized and rigorously annotated throughout the temporal sequences.

The VFSS-5k dataset contains over 5,000 accurate mask annotations extracted from dynamic videofluoroscopy sequences[10]. The dataset includes annotations for 10 distinct anatomical structures, specifically including the C1 through C7 cervical vertebrae, the pharynx, and mandibles[10]. These annotations take the form of pixel-wise segmentation masks, highly analogous to the mask annotations used for catheter shape detection in the robotic heart phantom studies.

The algorithmic utility of the VFSS-5k dataset is profound. Recent analyses utilizing medical foundation models have demonstrated high efficacy on this dataset[10]. Because fluoroscopic video is inherently plagued by speckle noise, patient movement, electronic noise, and low contrast-to-noise ratios, models trained on VFSS-5k learn to resolve spatial ambiguities across sequential frames[12]. Advanced frameworks leverage the Segment Anything Model (SAM), utilizing Vision Transformer (ViT) encoders pre-trained with masked autoencoders (MAE)[13]. These models utilize a dual-task optimization strategy during training: a main task for box-prompt segmentation utilizing a combination of Dice Loss and Binary Cross-Entropy (BCE) Loss, and an auxiliary task for point-prompt segmentation[10]. Through this rigorous training, models evaluating the VFSS-5k dataset achieve an average Dice coefficient of 0.868 across all anatomies[10]. For a researcher seeking to track a robotic instrument navigating the cervical spine, the temporal vertebral masks in VFSS-5k provide an ideal background suppression and anatomical localization training ground.

Quantitative Fluoroscopy and Kinematic Motion Datasets

In addition to swallow studies, continuous fluoroscopy is heavily utilized in biomechanical research to measure in vivo intervertebral motion. These datasets are essential for understanding how the spine deforms under load, which is a critical variable when a robotic

system is applying force to a vertebra during drilling or screw placement.
The Reference Database of Continuous Vertebral Flexion and Return is a prime example of this modality[15]. This dataset focuses on lumbar spine kinematics (L1 to S1) and captures standing flexion and return motions, quantitatively assessing intervertebral rotation and translation[15]. During extreme flexion, specific vertebrae, such as L5-S1, exhibit anti-directional motion in their outward path before returning to their starting position[16]. While these kinematic datasets may lack the dense pixel-wise masks of VFSS-5k, they provide raw, unoccluded dynamic sequences of the lumbar spine moving through 3D space. They offer continuous temporal data ideal for training unsupervised or self-supervised optical flow networks designed to track vertebral motion during patient respiration or surgical manipulation.

Dataset Name	Imaging Modality	Dataset Scale	Primary Annotation Format	Target Anatomy
VFSS-5k	Videofluoroscopy	5,000 labeled frames	Pixel-wise Segmentation Masks	Cervical Spine (C1-C7), Pharynx
QF Kinematics	Dynamic Fluoroscopy	Continuous sequences	Kinematic motion plots	Lumbar Spine (L1-S1)
VinDr-SpineXR	Static 2D Radiography	10,466 images	Bounding Boxes (13 classes)	Full Spine, Pathological Lesions
CSXA	Static 2D Radiography	4,963 images	23 Keypoints (Landmarks)	Cervical Spine (C2-C7)
CTSpine1K	3D Computed Tomography	1,005 volumes	3D Voxel Segmentation	Full Spine (Various conditions)

Simulated Multi-View Intraoperative Fluoroscopy and 3D-2D Registration

A profound limitation in the development of machine learning for robotic surgery is the extreme scarcity of paired 3D volumetric data and corresponding 2D intraoperative projection data. To circumvent this bottleneck, the prevailing methodology involves utilizing large-scale Computed Tomography (CT) datasets to generate Digitally Reconstructed Radiographs (DRRs). This

mathematical simulation perfectly replicates the exact physical conditions of a C-arm fluoroscope, yielding limitless, perfectly annotated multi-view fluoroscopy datasets.

The X23D Fluoroscopy Reconstruction Pipeline

The most advanced implementation of this sim-to-real paradigm is the X23D project, developed by researchers at the Balgrist University Hospital and ETH Zurich[2]. The X23D system is an artificial intelligence pipeline designed to intraoperatively estimate the 3D shape of a patient's lumbar vertebrae directly from sparse, multi-view X-ray data—typically utilizing only three to four fluoroscopic shots[2].

To generate the foundational training data, the researchers utilized the public CTSpine1K dataset. A custom software framework was engineered to pass these 3D volumes through a physical rendering equation, integrating the X-ray attenuation map along the ray initiating from the simulated source to the image plane[2]. This synthesis generated a massive fluoroscopic dataset containing 17,420 simulated fluoroscopy images[2]. Crucially, because the DRRs are generated from pre-segmented 3D CT volumes, the dataset automatically and flawlessly yields several critical annotations.

First, the dataset provides exact Projection Matrices (P-Matrices), which detail the precise intrinsic and extrinsic camera parameters, including focal length, source-to-detector distance, and spatial rotation[2]. Second, it outputs 2D vertebral bounding boxes formatted according to the standard COCO dataset specifications[2]. Finally, it generates 2D pixel-wise vertebral segmentation masks that perfectly identify the cortical and cancellous bone against the surrounding background[2].

The X23D dataset represents the absolute closest structural match to the robotic cardiac sheath dataset. It provides multi-view fluoroscopy that mimics bi-plane geometry, corresponding mask annotations for full shape detection, and is explicitly engineered for surgical navigation and pedicle screw placement. The inclusion of P-matrices allows for epipolar-constrained geometry, directly mirroring the mathematical foundation used for the 3D reconstruction of large-diameter transcatheter sheaths[1].

The LevelCheck Registration System

Parallel to the development of X23D, the LevelCheck system was engineered to provide real-time decision support in spine surgery by automatically labeling vertebral levels in intraoperative fluoroscopy using GPU-accelerated, intensity-based 3D-to-2D registration[18].

The dataset underlying LevelCheck is massive, consisting of 50,000 simulated fluoroscopic images derived from 10 high-resolution patient CT datasets[18]. The generation process programmatically varied the C-arm poses—including complex translations, rotations, and scaling—to mathematically approximate the natural variability of a human operator positioning a C-arm in a chaotic operating room environment[18]. The coordinate systems were rigorously defined: the C-arm coordinate system was defined at the isocenter, with axes aligned to the detector, while the preoperative CT coordinate system was defined at the center of the CT volume, aligned with the patient's anatomical axes[18].

By defining a nominal anterior-posterior (AP) position and introducing random geometric perturbations, the LevelCheck dataset rigorously evaluates the sensitivity of registration

algorithms to geometric calibration errors. Researchers discovered that adding a ± 300 mm translational offset to the calibration parameters allowed them to test if their algorithms could still compute the projected target position based purely on the relative relationships of surrounding tissues[19]. For robotic tracking, datasets like LevelCheck prove that robust algorithms can maintain accuracy even when the physical calibration of the fluoroscopy equipment degrades during a procedure.

Bi-Plane Kinematics and Experimental Robotic Tracking Datasets

The cardiac dataset referenced in the primary query utilized an explicitly controlled robotic system in conjunction with a bi-plane fluoroscopy setup. Finding public datasets that combine physical robotics, tissue phantoms or cadavers, and true bi-plane X-ray acquisition requires examining advanced orthopedic kinematic studies.

The Veriserum Bi-Plane Dataset

While focused on the femur and tibia rather than the vertebrae, the Veriserum dataset is methodologically identical to the experimental setup utilized by Wang et al. (2025) for robotic sheath delivery[1]. Veriserum is an open-source dataset explicitly designed to benchmark deep learning registration algorithms for dual-plane fluoroscopic analysis[1].

The imaging system employed for Veriserum comprised two X-ray tubes, two collimators, and two X-ray image intensifiers connected to high-speed cameras, all constructed in a precise bi-planar configuration[1]. The two X-ray planes were mounted at an angle of $110^\circ \pm 0.5^\circ$, featuring source-to-intensifier distances of 1850 mm and 1855 mm[1]. The image acquisition was performed at a tube voltage of 40 kV, yielding fluoroscopic images with a resolution of 1664 × 1600 pixels[1]. This resolution significantly exceeds the 800×800 parameter of the referenced cardiac dataset, providing exceptional clarity for algorithmic training.

To generate the data, surgical implants were securely mounted to the end-effector of an industrial-grade six-axis precision robot (HORST600), operating with a documented precision of ± 0.05 mm[1]. The robotic system sequentially moved the implants through predefined kinematic trajectories, holding each position for 500 milliseconds to ensure the end-effector remained perfectly stationary during X-ray acquisition, thereby eliminating motion blur[1]. Furthermore, the dataset includes raw calibration phantom measurements and non-linear calibration functions[1]. This critical inclusion allows algorithms to calculate exact source-to-intensifier distances and mathematically correct for the spherical distortion that is inherently produced by traditional X-ray image intensifiers[1]. For researchers developing algorithms for robotic guide sheaths or spinal drills, applying their models to the Veriserum dataset validates the mathematical core of epipolar geometry, camera calibration, and 3D pose estimation, regardless of the specific underlying bone morphology.

SpineDepth and SpineAlign: Multi-Modal Registration

Moving closer to actual spinal instrumentation, datasets evaluating pedicle screw insertion offer

profound insights into the challenges of robotic navigation in dense bone.
The SpineDepth dataset is an ex-vivo collection consisting of RGB-D (color and depth) recordings of pedicle screw placement performed on ten cadaveric specimens[8]. For each specimen, preoperative 3D vertebral models extracted from CT scans are co-registered with the intraoperative field[8]. Ground-truth vertebral poses relative to the RGB-D sensor are provided for every frame via high-precision optical tracking[8]. Uniquely, the dataset evaluates frames with up to 50% occlusion, perfectly mimicking realistic intraoperative scenarios where the surgeon's hands or robotic arms block the camera's view[8].
Complementing this is the SpineAlign dataset, an in-vivo collection containing data from 24 lumbar open spine surgeries[8]. This dataset provides coarse alignment between preoperative CT or MRI point clouds and the intraoperative data[8]. To process these complex datasets, state-of-the-art segmentation modules utilize KPConv in a U-Net-like encoder-decoder architecture[8]. KPConv is specifically chosen for its ability to extract features from irregular and sparse point clouds. Following segmentation, a GeoTransformer module predicts the rigid transformation required to align the source and target point clouds, focusing strictly on regions identified by binary masks as relevant anatomical bone[8]. These datasets parallel the cardiovascular tracking task by focusing on the precise localization of a rigid body relative to a dynamically shifting anatomical target.

Pedicle Screw Tracking and YOLO-OBB Segmentation

Fluoroscopic tracking of surgical tools, such as drills or pedicle screws, is the direct spinal equivalent to tracking a transcatheter guide sheath. Datasets focusing on pedicle screw pose estimation have rapidly emerged, utilizing advanced convolutional neural networks.
Recent deep learning frameworks have transitioned from standard Axis-Aligned Bounding Boxes (AABB) to Oriented Bounding Boxes (OBB) to capture the rotational trajectory of surgical implants[5]. For instance, a highly specialized dataset constructed from commercially available pedicle screws was trained using the YOLOv8-OBB architecture[5]. The annotations in this dataset predict the screw's center point (x, y), width (w), height (h), and crucially, the rotation angle (r)[5]. These networks achieve exceptional F1 scores of 0.86 in detecting the orientation and progression angle of the screw shaft within the C-arm's field of view[5].
More granular datasets isolate the input X-rays into three distinct mask classes: the screw head, the screw shaft, and the background tissue[22]. By extracting the screw shaft masks from bi-planar X-rays, the spatial correspondence between the two 2D projections can be established via epipolar geometry, enabling real-time 3D pose estimation of the surgical tool[22].

Pedicle Screw Parameter	Clinical Evaluation Standard	AI Detection Methodology	Accuracy / Output Metrics
Trajectory Angle	Trajectory deviation	YOLOv8-OBB	F1 Score: 0.86, mAP50[5]

	(degrees)	Rotation Angle (r)	
Medial/Lateral Breach	Gertzbein-Robbins Classification	GeoTransformer Point Cloud	Mean Euclidean Error: 1.04 mm[23]
Screw Shaft Position	Ravi Score	Multi-class CNN Masking	93% accuracy (Synthetic)[22]
Tip / Tail Localization	Zdichavsky Score	Bounding Box Center (x, y)	Tip error: 1.67 mm, Tail: 1.78 mm[23]

The clinical necessity of these high-precision datasets is underscored by the severe morbidity associated with misplaced pedicle screws. Accuracy is rigorously defined using clinical scales such as the Gertzbein-Robbins classification, where Grade 0 and 1 represent clinically accurate placement without neurovascular compromise, and Grades 2 and 3 represent dangerous cortical breaches[22]. Robotic navigation algorithms trained on these datasets must consistently achieve sub-millimeter accuracy to prevent medial breaches into the spinal canal or lateral breaches into surrounding vascular structures[22].

Large-Scale Static Spine Radiography Datasets for Pre-training

To effectively utilize dynamic fluoroscopy, machine learning models—whether based on ResNet, U-Net, or Vision Transformers—require robust feature extraction backbones. These backbones are typically pre-trained on massive static radiographic datasets. While these static datasets lack the temporal dimension, their sheer scale and highly detailed anatomical annotations are vital for establishing baseline spinal segmentation and localization capabilities.

The VinDr-SpineXR Dataset

The VinDr-SpineXR dataset is currently the largest annotated medical image dataset explicitly designed for spinal lesion detection and classification from radiographs[4].

The dataset is massive in scale, containing 10,466 spine X-ray images sourced from 5,000 distinct clinical studies[4]. To facilitate standardized machine learning evaluation, the data is split into 4,000 studies (8,389 images) designated for training, and 1,000 studies (2,077 images) strictly reserved for testing[4]. Experienced radiologists manually annotated the images using bounding boxes, covering 13 distinct types of abnormalities[4]. These classes include disc space narrowing, fractures, osteophytes, spondylolisthesis, and the presence of surgical implants or foreign bodies[4].

While VinDr-SpineXR utilizes bounding boxes rather than the dense pixel masks of the cardiovascular reference dataset, it remains foundational for training region-based object detection algorithms such as Faster R-CNN or YOLO. A model pre-trained on the

VinDr-SpineXR dataset learns to instantly recognize vertebral bodies and metallic implants despite heavy radiographic overlap and varying tissue densities—a skill that is effortlessly transferred to continuous fluoroscopic tracking tasks.

CSXA: The Cervical Spine X-ray Atlas

For highly precise anatomical localization, keypoint datasets offer an essential intermediate step between bounding boxes and full segmentation masks. The Cervical Spine X-ray Atlas (CSXA) fills a significant gap in high-quality, localized cervical imaging[6].

The CSXA consists of 4,963 raw PNG images and 4,963 corresponding JSON annotation files[6]. Each image in the dataset features exactly 23 precise keypoints[6]. The topology of these keypoints is designed to identify specific anatomical landmarks essential for surgical planning, including the central point and lower endplate of C2, and the four corner points plus the centroid for each vertebra from C3 through C7[6].

The quality control applied to the CSXA annotations is extraordinarily rigorous. The annotations underwent a three-tier cross-checking process performed by four orthopedic spine surgeons with an average of six years of clinical experience, ensuring sub-millimeter reliability across the entire dataset[6]. In robotic navigation, calculating the exact spatial orientation of a vertebra (pose estimation) is mathematically simpler and vastly less computationally intensive when using keypoints rather than parsing a full dense mask. Algorithms trained on CSXA can be rapidly adapted to real-time fluoroscopy to provide continuous, low-latency pose estimation of the cervical spine while a robotic drill approaches the surgical site.

Other complementary datasets include FracAtlas, which provides fracture classifications, bounding box localization, and segmentation masks for musculoskeletal radiographs[26]. Additionally, the BUU-LSPINE dataset provides 3,600 lumbar spine radiographs with annotations for individual vertebrae, commonly utilized alongside VinDr-SpineXR for benchmarking automated reporting and multi-label classification tasks[27].

Volumetric Spine Datasets for Fluoroscopic Synthesis

As demonstrated by the X23D pipeline, the modern approach to generating perfect fluoroscopic datasets requires high-resolution 3D source data. Several volumetric datasets serve as the foundation for these advanced simulation engines.

CTSpine1K and the VerSe Challenges

The most critical resource for generating spinal DRRs is CTSpine1K, a dataset curated to overcome the limitations of earlier, smaller collections[29]. CTSpine1K comprises 1,005 CT volumes, containing over 500,000 labeled slices[29]. The annotations include over 11,100 individually segmented vertebrae, covering diverse pathological appearances and varied scanner manufacturers[29].

This dataset builds upon the foundational work of the VerSe'19 and VerSe'20 (Large Scale Vertebrae Segmentation) MICCAI challenges. VerSe'19 provided 160 CT scans featuring 1,735 vertebrae, while VerSe'20 expanded this scope to 300 CT scans featuring atypical anatomies, such as transitional vertebrae and an anomalous L6 vertebra[29]. By applying complex ray-tracing algorithms to the dense voxel masks of CTSpine1K, researchers can generate customized

fluoroscopy datasets that perfectly mimic the 800x800 bi-plane setup of the robotic cardiac sheath experiments, complete with guaranteed ground-truth segmentation. Advanced deep learning models, such as the 3D full-resolution nnU-Net, are frequently deployed on these datasets to automate the baseline segmentation process, utilizing contracting paths for image downsampling and expansive paths for precise upsampling[30].

SPIDER: Lumbar Spine MRI Benchmarks

While X-ray attenuation is best simulated via CT, magnetic resonance imaging (MRI) datasets provide unparalleled soft-tissue contrast, which is critical for tracking neurological boundaries during surgery. The SPIDER dataset (Lumbar spine segmentation in MR images) serves as a multi-center public benchmark [cite: 2, 4-7].

SPIDER consists of 447 sagittal T1 and T2 MRI series acquired from 218 patients with a history of low back pain across four different hospitals[33]. The dataset provides manual reference segmentations of vertebrae, intervertebral discs (IVDs), and the spinal canal[33]. The labeling scheme is highly systematic: the lowest lumbar vertebra is assigned the label 1, progressing sequentially upward, while IVDs are labeled starting at 201, and the spinal canal is labeled 100[35]. Furthermore, it includes extensive clinical data, such as radiological gradings for Modic changes, disc herniation, narrowing, and Pfirrmann grades[33]. While not directly fluoroscopic, cross-modality registration—such as registering a real-time intraoperative fluoroscopic frame to a preoperative SPIDER MRI volume—is a highly active area of research in robotic spine surgery, ensuring that surgical instruments remain strictly outside the spinal canal[20].

Sim-to-Real Domain Adaptation and Next-Generation Volumetric Rendering

A critical insight derived from the analysis of simulated datasets like X23D is the persistent "domain gap" between simulated DRRs and real intraoperative fluoroscopy. DRRs, while geometrically perfect, fail to accurately replicate the polychromatic beam physics, scatter radiation, multi-material interactions, and detector-specific noise inherent to a physical C-arm[37]. Consequently, deep learning models trained exclusively on synthetic data frequently suffer catastrophic performance drops when deployed on real patients[37].

To bridge this gap, researchers have curated paired datasets combining synthetic DRRs and real ex-vivo fluoroscopic images captured from identical perspectives[37]. Utilizing Generative Adversarial Networks (GANs), specifically Pix2Pix architectures, these models are trained to perform sophisticated style transfer[38]. The GAN consists of a generator and a discriminator, optimized using a combination of adversarial loss and L1 loss to improve image quality and reduce blurring[38]. This process translates the noisy, low-contrast real-world fluoroscopy into the clean, standardized domain of the synthetic training data. For robotic interventional research, this implies that tracking algorithms trained on virtually generated mask annotations can be seamlessly and safely deployed in physical phantom or in-vivo robotic experiments.

Differentiable Gaussian Splatting and Statistical Shape Models

Recent advancements in differentiable rendering, specifically 3D Gaussian Splatting, are revolutionizing how 2D fluoroscopy datasets are utilized for spatial navigation. Traditional

CT-to-fluoroscopy registration relies on rigid voxel volumes, which struggle with deformed patient anatomies. Modern pipelines instead use Statistical Shape Models (SSM) parameterized by shape coefficients and rigid transformations[39].

In this architecture, Gaussian Splatting renders the 2D projections as elliptical blobs, accurately accounting for local surface orientation and perspective[39]. This creates an end-to-end differentiable pipeline, allowing gradient flow through both the shape and pose parameters[39]. By comparing these rendered projections to real X-rays using multi-view Vision Transformers (ViT) to calculate a learned similarity loss, researchers can optimize both the vertebral shape and pose simultaneously[39]. This technology eliminates the need to acquire hundreds of projections (as required by Cone-Beam CT), allowing accurate 3D reconstruction from a sparse set of arbitrary fluoroscopic views, drastically reducing patient radiation dose while maintaining the accuracy required for robotic alignment.

Synthesis and Strategic Recommendations

For a research team transitioning from cardiovascular robotic sheath delivery to spinal interventions, the acquisition, simulation, and integration of datasets must be highly strategic. The following synthesis provides a roadmap for developing a spinal navigation pipeline equivalent to the bi-plane cardiac studies.

1. **Temporal Tracking and Sequence Analysis:** To develop algorithms capable of tracking a robotic tool across noisy, sequential frames, researchers should utilize the VFSS-5k dataset[10]. Despite its clinical focus on dysphagia, it stands as the premier public dataset providing dynamic, continuous videofluoroscopy of the cervical spine with dense, pixel-wise mask annotations.
2. **Bi-Plane Calibration and Robotic Kinematics:** To mathematically validate epipolar 3D reconstruction, researchers should adopt the Veriserum dataset[1]. Its rigorous dual-plane fluoroscopy geometry (set precisely at 110 degrees), integration with industrial robotic end-effectors, and inclusion of distortion-correction calibration phantoms perfectly replicate the environment required for accurate stereotactic navigation.
3. **3D Reconstruction and Navigation Simulation:** To generate the massive quantities of data required to train deep learning pose estimators, researchers should implement the X23D pipeline methodology[2]. By utilizing the CTSpine1K volumetric dataset to procedurally generate custom Digitally Reconstructed Radiographs (DRRs), teams gain total control over virtual C-arm poses. This enables the simulation of any bi-plane or multi-view angle, automatically yielding P-matrices, COCO bounding boxes, and pixel-perfect ground-truth masks without the need for manual human annotation.
4. **Feature Extraction Pre-training:** To ensure neural networks possess robust foundational knowledge of spinal anatomy, convolutional or transformer-based backbones should be pre-trained on massive static repositories such as VinDr-SpineXR[4] and the CSXA[6]. This ensures that models learn to differentiate fundamental radiographic textures—such as cortical bone, osteophytes, and metallic implants—prior to fine-tuning on complex dynamic tracking tasks.

The landscape of spine-focused fluoroscopic datasets is fundamentally bifurcated into massive

static diagnostic archives and highly specialized, simulated kinematic models. By synthesizing 3D volumes into 2D projections and utilizing generative domain adaptation via GANs, researchers can successfully recreate the rigorous, mask-annotated, bi-plane environments required to safely pioneer the next generation of robotic spinal interventions.

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